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# Quantum Yield of Cyclobutane Pyrimidine Dimer Formation Via the Triplet Channel Determined by Photosensitization

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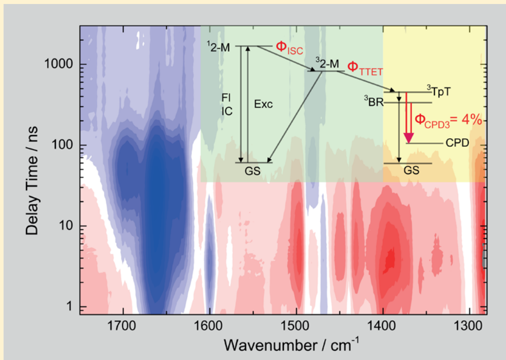
Supporting Information

**ABSTRACT:** UV-induced formation of the cyclobutane pyrimidine dimer (CPD) lesion is investigated by stationary and time-resolved photosensitization experiments. The photosensitizer 2′-methoxyacetophenone with high intersystem crossing efficiency and large absorption cross-section in the UV-A range was used. A diffusion controlled reaction model is presented. Time-resolved experiments confirmed the validity of the reaction model and provided information on the dynamics of the triplet sensitization process. With a series of concentration dependent stationary illumination experiments, we determined the quantum efficiency for CPD formation from the triplet state of the thymine dinucleotide TpT to be 4 ± 0.2%.




## ■ INTRODUCTION

Illumination by ultraviolet light is the most important threat to the integrity of genetic information on living organisms.[1,2] Among the different UV-induced photolesions, the cyclobutane pyrimidine dimer (CPD) between neighboring thymine (T) bases is the most frequent lesion and is thought to be a major cause of mutation and skin cancer.[3,4] The molecular reaction mechanisms leading to CPD formation have been the topic of intense discussions since the 1960s, when the CPD was identified as a major DNA photolesion.[5,6] For diluted monomeric thymine molecules irradiated by ultraviolet light, CPD formation is observed with small efficiency.[7] Here, lesion formation must be preceded by a diffusional collision of the excited thymine with another thymine molecule to allow dimer formation. Considering the long collision times, which are in the many nanosecond regime for typical concentrations of millimolar, CPD formation in solutions of monomeric T could only occur via the long-lived triplet state. However, low-temperature experiments performed in the 1960s also supported the idea that CPD formation should happen via an efficient singlet channel.[8] Very recently, a series of investigations using ultrafast infrared spectroscopy could show that upon UV-C illumination (at 266 nm) of single-stranded thymines the dominant fraction of CPD lesions is formed directly via the excited singlet state.[9] The formation time is less than 1 ps.[10] Banyasz et al. found that the contribution of triplet $^3\pi\pi^*$ states to CPD formation should be lower than 10%.[11] This finding was confirmed by time-resolved IR experiments on thymine single strands.[12] It was also stated that less than 15% of the excited triplet states react to the CPD lesion.[12]

In order to quantify CPD formation from the excited triplet state in the thymine dinucleotide TpT, we use selective triplet excitation via a sensitizing molecule,[13,14] which can transfer its triplet energy to TpT. The indirect excitation was necessary, hence the very low triplet yield upon direct optical excitation of the singlet $^1\pi\pi^*$ state prevents a quantitative analysis. Acetophenone, acetone, and benzophenone are often used in sensitization experiments.[15−19] The triplet states of acetophenone,[20] benzophenone,[21] and their interaction with DNA duplex[22−24] have been intensively studied. However, these molecules absorb very weakly in the UV-A and thus prevent time-resolved studies of the sensitization process without direct excitation of the nucleotides. The situation is even more complex for double-stranded DNA. For example, oligonucleotide duplex and cellular DNA show a weak but not vanishing absorption in the UV-A range, which may allow direct CPD formation.[25,26] In order to minimize the direct absorption by nucleotides in the present sensitization experiments on TpT we chose the sensitizing molecule 2′-methoxyacetophenone (2-M), which has a strong extinction coefficient at 320 nm ($\varepsilon$ = 2560 $M^{-1}$ $cm^{-1}$ for 2-M compared to ca. 10 $M^{-1}$ $cm^{-1}$ for TpT).[27] Additionally, the energy of the triplet state of 2-M is high enough for an efficient transfer of triplet excitation to a thymine dinucleotide.

In this paper, we start with a series of stationary experiments and we use the photosensitizer 2-M to determine the quantum yield of CPD formation via the triplet channel to be $\Phi_{CPD3} \approx$ 4%. Additional time-resolved experiments are performed to support the proposed reaction model and to quantify its important rates.

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## ■ MATERIALS AND METHODS

The 2-M was purchased from Sigma-Aldrich and thymidylyl-(3′ → 5′)-thymidine (TpT) was purchased from Metabion. Both chemicals were used without further purification. All measurements were performed in a deuterated water solution with 50 mM phosphate buffer at the concentrations given in the text. In all experiments with the sensitizer, a concentration of 2-M of 5 mM was used.

The stationary illumination for the determination of the quantum yield was performed by narrow-band ($\Delta\lambda \approx 0.1$ nm) nanosecond pulses from a laser-OPO system (NT242, Ekspla) operated at a repetition rate of 1 kHz (13 $\mu$J per pulse, elliptical illumination spot 3.5 mm × 10 mm). The sample circulation setup was composed of a flow cell (a Teflon spacer with a thickness of 108 $\mu$m, sandwiched by two $CaF_2$ windows) and a standard 4 mm × 10 mm cuvette with fused silica windows (in which the sample was stored and illuminated). They were connected with PTFE tubes and the sample was circulated with a peristaltic pump. The time-resolved experimental setup was based on a combination of a Ti-sapphire laser-amplifier system (Spitfire Pro, SpectraPhysics) to supply the IR probe pulse and the laser-OPO system for excitation. The Spitfire Pro emitted pulses with 100 fs duration at 800 nm and a repetition rate of 1 kHz. The probe light in the mid-IR from 1280 to 1750 $cm^{-1}$ is generated by a combination of a noncollinear and a collinear optical parametric amplifier, followed by difference frequency mixing in an $AgGaS_2$ crystal.[28] The transmitted probe pulse is spectrally dispersed (Chromex 250 IS, Bruker) and recorded with a 64 Channel MCT array detector (IR-0144, Infrared Systems Development). We combine four spectral ranges, which overlap at the transition range. The recorded spectra of each range are suitably scaled. The pump pulses are obtained from the NT242 with 3 ns duration and 1 kHz repetition rate. The pump energy was 6 $\mu$J for the time-resolved experiment with a spot diameter of ca. 120 $\mu$m. In order to avoid the molecular rotation artifacts, the polarizations of both pump and probe pulses were orientated to yield the magic-angle. The same flow cell as for the stationary measurements was used in which the sample solution is exchanged between two successive pump pulses. All experiments were performed at room temperature (22 °C) under normal atmosphere and ambient oxygen concentration. The transient absorption data and 2D absorbance change data of Figures 3 and 5 are corrected for the contribution of solvent heating remaining after the decay of the excited electronic state. The decay associated difference spectra (DADS) and species associated difference spectra (SADS) are fitted without any solvent heating subtraction.

## ■ THE REACTION SCHEME

In Figure 1, we present a combined reaction scheme for the description of sensitizer mediated CPD formation (left part) and direct CPD formation (right part). Upon optical excitation of the allowed $^1\pi\pi^*$ singlet transition of the dinucleotide TpT, ultrafast internal conversion (IC) dominates the decay of the excited electronic state $^1$TpT*. The corresponding time constant is in the subpicosecond range. Besides this fast internal conversion, there is direct CPD formation with the yield $\eta_{CPD1}$ and intersystem crossing (ISC) with the yield $\eta_{ISC}$. According to recent experiments of the oligomer $(dT)_{18}$, the triplet state ($^3$TpT) populated by intersystem crossing should decay on the time scale of ≈10 ns to a biradical state ($^3$BR) and from there to the ground state. Recent measurements have

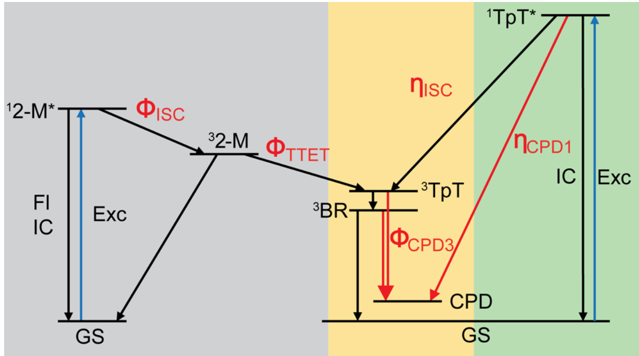


**Figure 1.** Reaction scheme for direct and sensitized formation of the CPD photolesion. After excitation of the allowed $\pi\pi^*$ transition, CPD formation may occur in an ultrafast reaction channel directly from $^1$TpT* with an efficiency $\eta_{CPD1}$ or after intersystem crossing (ISC) with an efficiency $\eta_{CPD3} = \eta_{ISC} \cdot \Phi_{CPD3}$. The sensitized reaction channel requires ISC of the sensitizer and TTET to the $^3$TpT prior to CPD formation. Its efficiency depends on the TpT concentration $C$(TpT).

shown that the formation efficiency of the CPD photolesion via the triplet channel is low.[11] For $(dT)_{18}$, it has been proven that only a small amount of the excited triplet states could lead to CPD formation, that is, $\Phi_{CPD3} < 15\%$.[12] Combining $\Phi_{CPD3}$ with the ISC yield $\eta_{ISC}$ from ref 11 one may estimate an upper limit for $\eta_{CPD3}$, the contribution of the triplet channel after direct excitation of $^1$TpT*. Assuming that TpT has the same ISC yield as TMP, one can deduce $\eta_{ISC} \approx 1.3\%$ (at 266 nm) from the literature[11] and obtain $\eta_{CPD3} = \eta_{ISC} \cdot \Phi_{CPD3} < 0.2\%$. In order to obtain a definite value for $\eta_{CPD3}$ instead of an upper limit, we performed a triplet sensitizing experiment (see left part of Figure 1). In this experiment, the triplet sensitizer (2-M) is excited by light at 320 nm in a solution containing TpT and the sensitizing molecule. At 320 nm, the absorption coefficient of TpT is so small that a direct excitation of TpT can be excluded. Only triplet−triplet energy transfer (TTET) via the sensitizer 2-M may excite TpT into its triplet state $^3$TpT. It is known that the optically excited singlet state $^1$2-M* decays predominantly via intersystem crossing. The decay time is ca. 700 ps and the quantum yield of intersystem crossing $\Phi_{ISC}$ is 97%.[27] In a neat water solution of 2-M (concentration: 5 mM) at ambient oxygen concentration a lifetime of the triplet state $^3$2-M of $\tau_0 = 1/Z_0 = 400$ ns was observed.[27] If the solution contains TpT, the triplet state of 2-M may be quenched by TTET upon collision with TpT (transfer rate $Z_1$). Thereby the triplet state $^3$TpT is populated. The transfer efficiency for TTET, $\Phi_{TTET}$, could be deduced from the decay rates $Z_0$ and $Z_1$, where $Z_1 = Z_1(C)$ depends on the concentration $C = C$(TpT) of the triplet acceptor TpT

$$\Phi_{TTET}(C) = \frac{Z_1(C)}{Z_1(C) + Z_0} \quad \text{with} \quad Z_1(C) = \kappa \cdot C = \kappa \cdot C(\text{TpT}) \tag{1}$$

The bimolecular collision constant $\kappa$ is related to the diffusion of the involved molecules.[29] After the population of the triplet state $^3$TpT by TTET, the formation of the CPD lesion occurs from this state with the yield $\Phi_{CPD3}$. In this paper, we do not distinguish whether the CPD lesion is directly formed from $^3$TpT or via the biradical $^3$BR. For the total CPD formation yield $\Phi_{CPD\text{-}Sens}$ upon excitation of the sensitizer one calculates

$$\Phi_{CPD-Sens} = \Phi_{CPD-Sens}(C) = \Phi_{ISC} \cdot \Phi_{TTET}(C) \cdot \Phi_{CPD3} \tag{2}$$


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Considering this reaction scheme, there are two possibilities to determine the yield $\Phi_{CPD3}$. (i) One can measure the dependence of the CPD formation yield $\Phi_{CPD\text{-}Sens}(C)$ for various TpT concentrations $C$ and use eqs 1 and 2 to determine the collision constant $\kappa$ and the product $\Phi_{ISC} \cdot \Phi_{CPD3}$. Because the yield $\Phi_{ISC}$ is known,[27] the quantity $\Phi_{CPD3}$ can be obtained directly. (ii) A second approach uses a time-resolved experiment to determine $\Phi_{TTET}(C)$ by measuring the triplet−triplet energy transfer rate $Z_1$ for a certain TpT concentration. A stationary measurement of $\Phi_{CPD\text{-}Sens}$ at the same concentration allows then to determine $\Phi_{CPD3}$. Furthermore, the time-resolved experiment gives independent information on the validity of the assumed reaction scheme.

## ■ RESULTS

**Quantum Yield Measurement with Stationary Spectroscopy.** In a first set of experiments, the formation of the CPD photolesion via the sensitizer 2-M was investigated by stationary infrared spectroscopy using different concentrations of TpT ranging from 2.29 to 18.32 mM. The concentration of 2-M was always 5 mM. Figure 2a shows the change of the IR absorbance spectra recorded after different illumination times for the concentration $C(\text{TpT})$ = 9.16 mM. With rising illumination times, corresponding to rising illumination doses, we find a bleach of the TpT bands between 1600 and 1700 cm$^{-1}$. At the same time, there is increased absorption in the range 1300 to 1500 cm$^{-1}$ at the positions of the marker bands characteristic for CPD formation. An analysis of the absorption changes recorded as a function of the absorbed illumination dose allows us to determine the quantum yield for CPD formation. The quantum yield is defined as the number of converted molecules per unit time divided by the number of absorbed photons per unit time.[30] Therefore, it can be expressed as

$$\Phi_{CPD\text{-}Sens} = \frac{V}{I}\frac{dC(\text{CPD})}{dt} = \frac{V}{I}\frac{1}{\Delta\varepsilon \cdot d}\frac{d\Delta A}{dt} \qquad (3)$$

Here, $V$ = 1.2 mL is the volume of the sample, $I = P \times 2.6725 \times 10^{-9}$ mol/(mW·s) is the number of absorbed photons of the illumination light (320 nm) per unit time, and $P$ is the power of absorbed photons. $dC(\text{CPD})/dt$ is the change of the CPD-concentration per time. One measures directly the change in absorbance $d\Delta A/dt$, which can be expressed with the Beer−Lambert law as the product of $dC(\text{CPD})/dt$, the change in extinction coefficient $\Delta\varepsilon$ = 343.7 /(M·cm) upon CPD formation at 1402 cm$^{-1}$ and the light path $d$ = 108 $\mu$m in the IR sample. The inset of Figure 2a shows the absorbance changes $\Delta A$ at 1631 and 1402 cm$^{-1}$ recorded as a function of the illumination time for constant irradiation power. At low illumination doses, the absorbance change $\Delta A$ can be fitted with a linear function;[31] the slope of the fitted curve $d\Delta A/dt$ is used for the calculation of the quantum yield $\Phi_{CPD\text{-}Sens}$. For a concentration of $C(\text{TpT})$ = 9.16 mM a value of $\Phi_{CPD\text{-}Sens}$ = 3.3 ± 0.2% is obtained using eq 3. Similarly we could get the quantum yield for other concentrations $C(\text{TpT})$ = 2.29, 4.58, and 18.32 mM. The experimental results from a set of independent measurements at each concentration are plotted in Figure 2b. In an additional experiment (9.16 mM TpT, 5 mM 2-M), we evaluated a potential influence of the oxygen concentration on the quantum yield of the investigated solutions. As shown in the Supporting Information (Figure S5), a reduction of the oxygen concentration by a factor of 3 from the otherwise used ambient conditions did not change the reaction yield. This excludes the role of quenching by oxygen under the experimental conditions used.

As can be seen from eq 2, the quantum yield depends on the concentration of TpT acceptor molecules. A ready combination of eqs 1−3 leads to an equation that deduces the unknown model parameters from a plot of the experimental data

$$\frac{1}{\frac{d\Delta A}{dt}} = a + a \cdot \frac{Z_0}{\kappa}\frac{1}{C(\text{TpT})} \quad \text{with} \quad a = \frac{V}{I}\frac{1}{\Delta\varepsilon \cdot d}\frac{1}{\Phi_{ISC} \cdot \Phi_{CPD3}} \qquad (4)$$

A linear fit of the experimental data according to eq 4 yields $a$ = 2.5 × 10$^5$ s and $(a \cdot Z_0/\kappa)$ = 3.9 × 10$^2$ mM·s. Using $1/Z_0$ = 400 ns, we determine the bimolecular collision constant $\kappa$ = 1.6 × 10$^9$ (M·s)$^{-1}$. The statistical variations of the individual measurements indicate that $\kappa$ should be between 1.4 × 10$^9$ (M·s)$^{-1}$ and 2.0 × 10$^9$ (M·s)$^{-1}$. Corresponding curves are plotted in Figure 2b. These values of $\kappa$ are in good agreement with published data for similar molecules.[16] We deduce $\Phi_{CPD3}$ = 4.1 ± 0.2% for the quantum yield of CPD formation via the triplet channel.

**Characterization of TpT Triplet State with Time-Resolved Spectroscopy.** Figure 1 shows that the triplet state $^3$TpT plays an important role in the whole reaction scheme. In order to characterize this state, especially its decay path and the decay rates, time-resolved experiments on TpT are performed. Although TpT possesses strong absorption bands in

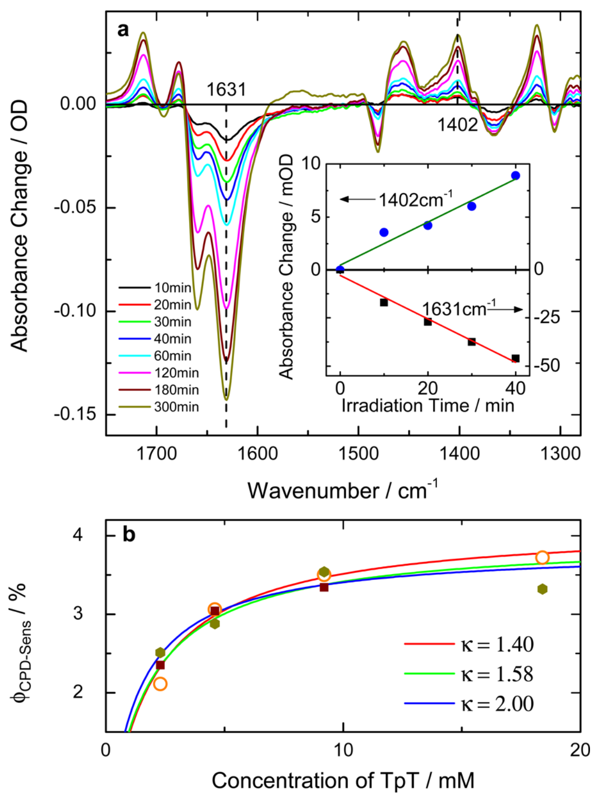


**Figure 2.** (a) Absorbance changes in the mid-IR of a solution of TpT (9.16 mM) and 2-M (5 mM) induced by illumination of the sensitizer 2-M at 320 nm after different irradiation times (absorbed irradiation 0.78 J/min). The rise of the CPD photodamage is well recognized at the CPD marker bands between 1500 and 1300 cm$^{-1}$.The rise of the 1402 cm$^{-1}$ marker band and the bleach of the original TpT (1631 cm$^{-1}$ band) are displayed in the inset. (b) Quantum yield $\Phi_{CPD\text{-}Sens}$ determined for different TpT concentrations (points) and a modeling of the yield (solid curves) for different values of $\kappa$.



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the UV-C to UV-B range, the triplet state can only be occupied via intersystem crossing (ISC) from the excited singlet state. From literature,[11] we know that the ISC efficiency of thymidine monophosphate (TMP) is low and decreases by increasing the excitation wavelength from 250 to 280 nm. At 250 nm, the ISC yield is about 4%. For the investigation of the reaction dynamics, we used a neat TpT solution (10 mM per thymine base) with 250 nm excitation pulses.

Absorbance changes in the mid-infrared from 1750 to 1280 $cm^{-1}$ and in the time range from 0.8 ns to 3 $\mu$s are shown in Figure 3a. No significant absorption dynamics were observed at later times (3 to 30 $\mu$s, data not shown). The transient absorption data of Figure 3 are corrected for the contribution of solvent heating remaining after the decay of the excited electronic state. Representative difference spectra recorded at indicated delay times are plotted in Figure 3b−e. The curves show that significant reaction dynamics occur in the time range up to 100 ns. Considering the duration of the exciting laser pulse of $t_p$ = 3 ns and the short lifetime of the excited $\pi\pi^*$ state of TpT (<1 ps[9]) we cannot resolve the ISC process from $^1$TpT* to $^3$TpT. Thus, in the present nanosecond experiment the triplet state $^3$TpT appears to be instantly occupied upon excitation. At early times, one observes the bleach of the ground-state (1700−1630 $cm^{-1}$) and some increased absorption between 1550 and 1300 $cm^{-1}$. From recent experiments on single-stranded thymine oligomers $(dT)_{18}$,[12] we know that the decay of the triplet state occurs within the first 100 ns with two transient species: the triplet state $^3$TpT and a biradical state $^3$BR. In our experiments on the dinucleotide TpT, we also observe kinetics in the same time range as can be seen from the pronounced differences of the transient spectra recorded at 10, 50, and 100 ns and at the end of the observation period (see Figure 3b−e). The corresponding decay time constants ($\tau_1$ = 22.5 ns and $\tau_2$ = 62 ns) were determined with a global fit using a sum of two exponential functions convoluted with the instrumental response function and an offset. The wavelength dependent fit amplitudes corresponding to the different time constants are called DADS. They are given in the Supporting Information in Figure S1. The time constants observed in the triplet decay of TpT are similar but somewhat longer than those observed for $(dT)_{18}$, where values of ∼10 and ∼60 ns were found.[12] For the assignment of the two kinetic components to molecular states, we may refer to the amplitude spectra (DADS) and calculate the difference spectra related to the intermediate states, that is, the SADS.[32] Figure 4 shows the SADS (red curves) calculated for the assumed sequential reaction model where $^3$TpT decays with $\tau_1$ = 22.5 ns to the biradical state $^3$BR which decays with $\tau_2$ = 62 ns to the ground state. These spectra exhibit the features known for the respective states from the experiments on $(dT)_{18}$. A direct comparison of the SADS of $(dT)_{18}$ and TpT is presented in the Supporting Information (S4). The $^3$TpT state shows the characteristic triplet band at 1600 $cm^{-1}$ and the bleach of the C═O bands in the range from 1700 to 1650 $cm^{-1}$. The $^3$BR state displays distinct bands between 1700 and 1600 $cm^{-1}$. The bleach of the 1630 $cm^{-1}$ band is reduced and a positive band appears around 1620 $cm^{-1}$, while the 1600 $cm^{-1}$ triplet band vanishes. The recorded spectra are fully consistent with a decay of the triplet state to the ground state TpT. However, the formation of a small amount of CPD lesions in this time range cannot be excluded. Even if it is present, its amplitude is too small to be detected in our experiment. The results on $^3$TpT

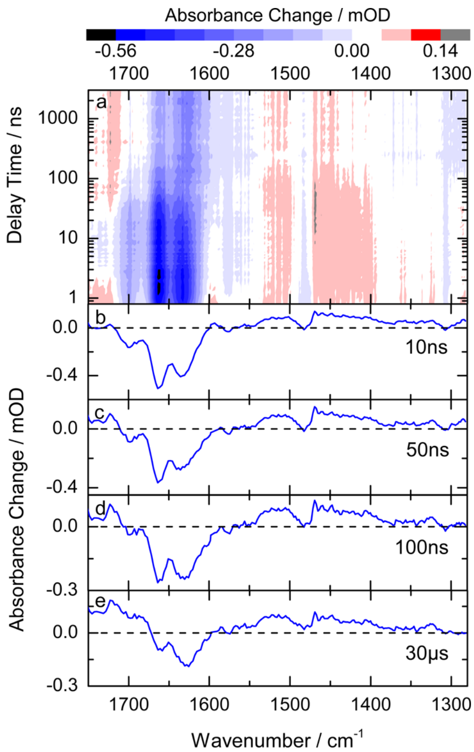


**Figure 3.** Transient absorption data for a TpT solution excited with nanosecond pulses at 250 nm. (a) Absorbance changes recorded as a function of probing wavenumber and delay time in color coding. Transient absorbance difference spectra recorded at delay times of 10 ns (b), 50 ns (c), 100 ns (d) and at the end of the observation range at 30 $\mu$s (e).

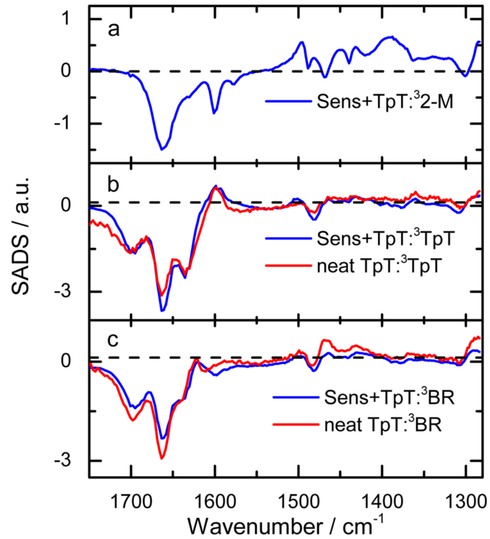


**Figure 4.** Species associated absorption difference spectra (SADS) calculated from the absorption difference data for neat TpT ($\lambda_{exc}$ = 250 nm, red curves) and for the sensitizing experiment (solution containing the sensitizer 2-M (5 mM) and TpT (18.32 mM) with $\lambda_{exc}$ = 320 nm, blue curves. SADS corresponding (a) to the triplet state $^3$2-M of the sensitizer, (b) to the triplet state $^3$TpT, and (c) to the biradical state $^3$BR. The reaction scheme presented in eq 5 was used for the calculation.

and $^3$BR will be used in the following data evaluation of the sensitizing experiment where these states are also present.

**The Sensitizing Experiment.** Independent information on the dynamics of the sensitizing reaction and an experimental support for the reaction scheme in Figure 1 are obtained from time-resolved experiments on a solution containing TpT at a high concentration (18.32 mM) and the sensitizer 2-M (5 mM). After illumination of the solution with nanosecond pulses at 320 nm, we find absorption transients on the time scale of few hundred nanoseconds (see Figure 5a). Traces for the absorbance changes recorded at 10 ns, 50 ns, 100 ns and at late times are displayed in Figure 5b–e. These absorbance changes differ significantly from those of neat TpT upon 250 nm excitation. Immediately after excitation (at 10 ns), we observe the bleach of the sensitizer bands at 1663 and 1600 cm$^{-1}$ combined with induced absorption due to the triplet bands of 2-M at 1500, 1450, 1430 and 1400 cm$^{-1}$. These spectral features clearly point to the presence of the triplet state $^3$2-M of the sensitizer.[27] The triplet bands of 2-M decay on the 30 ns time scale. At 50 ns, the spectrum shows indications known from the triplet state $^3$TpT. The spectrum resembles the difference spectrum of the triplet and the biradical state of $(dT)_{18}$ discussed in ref 12. At late times (50 $\mu$s), the spectra contain features that are known from the formation of CPD lesions (see Figure 2a and literature[10]). Figure 6 focuses on the time dependences. It compares the reaction dynamics of a neat sensitizer solution (black curve) and a solution containing both TpT and 2-M (red curve) recorded at 1663 cm$^{-1}$, that is, in the range of ground-state absorption of both TpT and 2-M. For

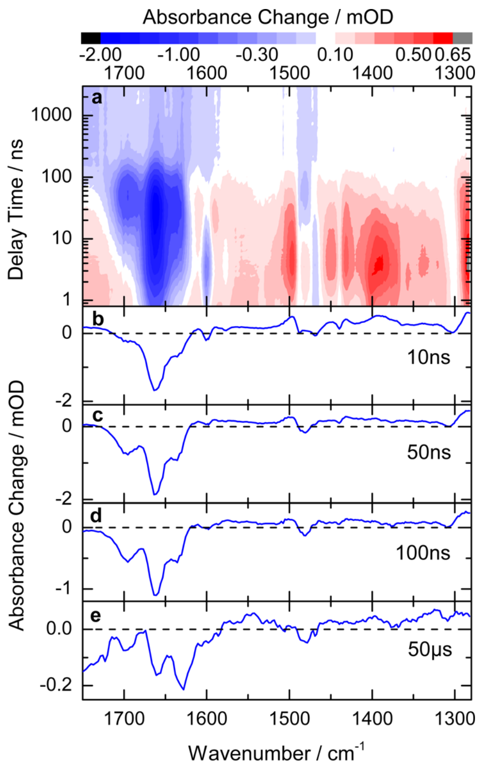


Figure 5. Transient absorption data for the sensitizing reaction (solution containing 2-M (5 mM) and TpT (18.32 mM)), excited with nanosecond pulses at 320 nm. (a) Absorbance changes recorded as a function of probing wavenumber and delay time in color coding. Absorption difference spectra recorded at delay times of 10 ns (b), 50 ns (c), 100 ns (d), and at the end of the observation range at 50 $\mu$s (e).

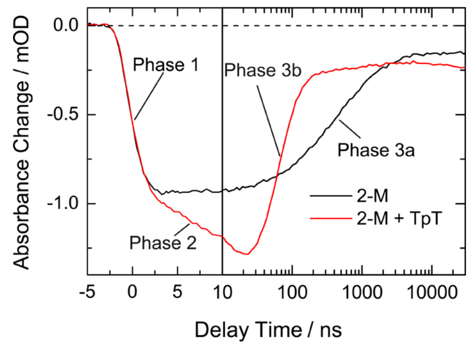


Figure 6. Time-dependent absorbance changes for neat 2-M (black trace) and the sensitizing solution containing 2-M and TpT (red trace) recorded at 1663 cm$^{-1}$. While the rise of the absorption bleach upon excitation and formation of $^3$2-M is similar in both samples (Phase 1), the pure 2-M solution shows a late absorption recovery with a time constant of 400 ns (Phase 3a). For the mixture, there is a secondary absorption bleach up to 30 ns (Phase 2) due to TTET to TpT before the absorption recovers with a time constant of 60 ns (Phase 3b).

neat 2-M, a rapid bleach points to the excitation of 2-M (phase 1). At late times, there is a decay, which can be fitted by a time constant of ca. 400 ns (phase 3a). The time constant and the transient spectra (see above) indicate that we have observed the excited triplet state $^3$2-M over the whole time range. The solution containing both 2-M and TpT displays at early times the same bleach (phase 1). However, the bleach increases further on the 10 ns scale and reaches its maximum at 25 ns (phase 2). This transient directly supports the idea of a triplet–triplet energy transfer (TTET) from $^3$2-M* to $^3$TpT. The latter decays subsequently with a time constant of ca. 60 ns (phase 3b) back to the ground state.

## DISCUSSION

**Modeling of the Time-Resolved Data.** For a more detailed analysis, the experimental data presented in Figure 5 are modeled by a sum of exponential functions. The obtained time constants and decay associated difference spectra (DADS) are evaluated within the scope of the reaction scheme shown in Figure 1. The collision constant $\kappa = 1.6 \times 10^9$ (M·s)$^{-1}$ obtained in the stationary experiments gives an estimate of the TTET time at the concentration $C(\mathrm{TpT})$ = 18.32 mM used in the time-resolved experiment: $\tau_{\mathrm{TTET}} = 1/Z_1(18.32\ \mathrm{mM}) = 1/(\kappa \cdot 18.32\ \mathrm{mM}) \approx 34$ ns. With this time constant, the lifetime for the state $^3$2-M is estimated: $\tau_{3\text{-}2\mathrm{M}} = 1/(Z_1(18.32\ \mathrm{mM}) + Z_0) \approx 31$ ns. Unfortunately this time constant is in the same range as the time constants of the decay of the triplet state $^3$TpT ($\tau_1$ = 22.5 ns) and $^3$BR ($\tau_2$ = 62 ns). A simple fit of the experimental data according to the reaction model would require three unrestricted time constants between 10 and 100 ns. Because such a fit is inherently mathematically unstable, we predetermine the time constants $\tau_1$ and $\tau_2$ from the results of the 250 nm experiment. $\tau_{3\text{-}2\mathrm{M}}$ is deduced from the time-dependent absorption data of Figure 5 as the only free time constant of the fitting procedure. The fit gives $\tau_{3\text{-}2\mathrm{M}} = 33 \pm 10$ ns. Using this value, we obtain a similar time constant for the TTET as estimated above, $\tau_{\mathrm{TTET}} = 1/(1/\tau_{3\text{-}2\mathrm{M}} - Z_0) \approx 36$ ns. The related decay associated spectra of all kinetic components ($\tau_{3\text{-}2\mathrm{M}}$, $\tau_1$, $\tau_2$, and offset) are presented in the Supporting Information (S2). We calculate the absorption difference spectra of the intermediate states (SADS) using the DADS and the central part of the reaction scheme of Figure 1.


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$$\text{2-M} \xrightarrow{h\nu,\text{ISC}} {}^{3}\text{2-M} \begin{cases} \xrightarrow[36\text{ ns}]{\tau_{\text{TTET}}} \begin{cases} {}^{3}\text{TpT} \xrightarrow[22.5\text{ ns}]{\tau_1} {}^{3}\text{BR} \xrightarrow[62\text{ ns}]{\tau_2} \text{TpT, CPD} \\ \text{2-M} \end{cases} \\ \xrightarrow[400\text{ ns}]{1/Z_0} \text{2-M} \end{cases} \tag{5}$$

The SADS of the sensitizing experiment are plotted in Figure 4 (blue curves) together with the SADS of the reference experiment on TpT (red curves). The comparison of the different SADS for $^3$TpT and $^3$BR shows a nice coincidence of the most important features and strongly supports the reaction model and the data evaluation procedure. Experiments with a lower concentration of C(TpT) (see Supporting Information (S3)) give consistent results with a somewhat reduced precision. From the time constant $\tau_{3\text{-}2M}$ = 33 ns obtained above we can calculate the collision constant $\kappa = 1/(\tau_{\text{TTET}}\, C(\text{TpT})) = (1/\tau_{3\text{-}2M} - Z_0)/C(\text{TpT}) = 1.5 \times 10^9\ (\text{M}\cdot\text{s})^{-1}$. It is in the same range as $\kappa = 1.6 \times 10^9\ (\text{M}\cdot\text{s})^{-1}$ found in the stationary experiments (see above). The question whether the CPD is formed from $^3$TpT or $^3$BR or from both remains open and cannot be answered unambiguously from the evaluation of the experimental results. The ambiguity is due to the presence of three time constants in the 10 to 100 ns range, the precision of the experimental data and the low reaction yield of about 4%. We will address this point below.

**CPD Formation: Triplet versus Singlet Channel.** The determination of the reaction yield $\Phi_{\text{CPD3}} \approx 4\%$ allows us to estimate the quantum yield $\eta_{\text{CPD3}}$ for CPD formation via the triplet channel after direct excitation with UV-C light. From the relation $\eta_{\text{CPD3}} = \eta_{\text{ISC}} \cdot \Phi_{\text{CPD3}}$ with $\Phi_{\text{CPD3}}$ determined above and $\eta_{\text{ISC}}$ from literature[11] one may quantify the relative efficiency of the singlet and the triplet channel for the dinucleotide TpT. For an excitation at 266 nm, the triplet yield for TMP was deduced from the literature to be $\eta_{\text{ISC}} \approx 1.3\%$.[11] In the Supporting Information (S6) we show that the triplet efficiencies are the same for the monomer TMP and the oligonucleotide $(\text{dT})_{18}$ (excitation wavelength 250 nm). Apparently the stacking of the thymine bases, which is more prominent in $(\text{dT})_{18}$ than in TpT, does not influence the triplet yield. Therefore, we use the same yield $\eta_{\text{ISC}} \approx 1.3\%$ also for TpT. With this value, one finds that only $\eta_{\text{CPD3}} \approx 1.3\% \cdot 4\% \approx 0.05\%$ of the absorbed ultraviolet photons at 266 nm generate the CPD photolesion via the triplet channel, whereas $\eta_{\text{CPD1}} = 1.5\%$[9] of the absorbed photons form the CPD lesion directly from the singlet state. Thus, at 266 nm CPD formation via the singlet channel is 30 times more efficient than via the triplet channel. The relative importance of the singlet channel is increased for illumination at longer wavelengths where the yield of triplet formation becomes smaller while the yield of CPD formation does not change.[11] As a consequence, direct absorption of the solar light, which reaches the earth's surface with reasonable fluence only at longer wavelengths, $\lambda > 290$ nm, generates the CPD lesion essentially via the singlet channel. However, sensitizing processes populating the triplet state after absorption of light in the range above 300 nm may be of major importance due to the significant CPD formation yield $\Phi_{\text{CPD3}} = 4\%$ and the high fluence of solar radiation at these wavelengths.

**The Molecular Processes of CPD Formation.** Direct CPD formation from the excited singlet state is faster than 1 ps.[10] This process occurs on a time scale where the relative arrangement of adjacent nucleobases in a strand is essentially frozen. It has been shown by molecular dynamics and quantum chemical modeling that only those molecules that are suitably prearranged in the moment of light absorption are able to undergo CPD formation within the lifetime of the excited singlet state.[33,34] The efficiency $\eta_{\text{CPD1}} = 1.5\%$ is determined by the probability of the reactive arrangement of the nucleotides whereas energetic constrains are relaxed due to the large energy excess of the singlet state relative to the CPD. The situation is different for the triplet channel. With a typical lifetime in the 10 ns range, the triplet states allow the nucleotides to test the complete conformation space of arrangements. Especially for the $^3$BR state, where one bond of the CPD is preformed, the conformational criteria are relaxed and should not limit the reaction yield. The finding that the CPD formation yield from the excited triplet state $\Phi_{\text{CPD3}} = 4\%$ is still rather low supports the idea that the energetic constrains determine the transfer efficiency.

In conclusion, the present investigation has quantified the yield of CPD formation via the triplet channel. With stationary and time-resolved sensitizing experiments using a modified acetophenone as a sensitizer, it was shown that the triplet channel is able to form the CPD with an efficiency of $\Phi_{\text{CPD3}} \approx 4\%$, presumably from the biradical state $^3$BR. The formation of the CPD lesion after the absorption of a photon at 266 nm into the $\pi\pi^*$ state of the TpT occurs 30 times more often via the singlet channel than via the triplet channel. However, the indirect sensitizing reaction with the abundant light in the UV-B and the UV-A range may be of major importance due to the large yield $\Phi_{\text{CPD3}} \approx 4\%$ in surroundings with suitable sensitizing molecules.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcb.5b08568.

Decay associated difference spectra (DADS), a comparison between species associated difference spectra (SADS) of TpT and $(\text{dT})_{18}$, transient absorption data for a sensitization measurement with a low TpT concentration, a quantum yield measurement at reduced oxygen concentration, and a comparison of triplet yields in TMP and $(\text{dT})_{18}$. (PDF)

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### Notes

The authors declare no competing financial interest.

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